The race engineer for the drivers who *don't have one*.

AI race engineer purpose-built for adaptive racers, veteran-team drivers, and grassroots competitors. Built on the IBM Granite stack.

apex-one-black.vercel.app · github.com/StephenSook/apex · Apache 2.0

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A race engineer is a *luxury good*.

£400-500

PER DAY · PROFESSIONAL RACE ENGINEER

Every Formula One driver has one. Most adaptive racers, veteran-team drivers, and grassroots competitors do not. The FIA lifted its single-seater ban on disabled drivers in December 2017. The barrier stopped being regulatory. It became economic.

> Standard AI race-engineer tools assume able-bodied physics. They encode throttle times brake equals zero. Penalizing simultaneity misdiagnoses every adaptive driver whose Certificate of Adaptations permits it through approved hand-control hardware.

APEX changes that. The Certificate of Adaptations is read at tensor level + routes the constraint per-driver. Same coaching pipeline says different things for different drivers based on what their COA says they are allowed to do.

Sarah Reynolds. Lap 17 of 19. *Two tenths off her PB.*

DRIVER · CIRCUIT · LAP

Britcar Trophy 2026. Donington Park GP. Lap 17 of 19 of qualifying. Time bleeding out in Sector 2 at Old Hairpin.

DEBRIEF

“lost the rears mid Old Hairpin again, can't trail-brake on the lever the way she did at Croft last month.”

FICTIONAL PERSONA by design. No real adaptive-driver identity. The hand-control hardware spec, the COA structure, and the lap-and-symptom debrief are derived from publicly documented Britcar regulations + the MME Motorsport electronic hand-control system (consent receipt 2026-05-22 from MME Motorsport d.o.o.).

HARDWARE

BMW M240i, MME Motorsport electronic hand-controls, COA section 3(c) approves dual-stage trigger permitting simultaneous brake-throttle actuation through corner entry.

PhysicsTTM. Three layers, composed.

LAYER 1 · FORECASTER

Granite TimeSeries TTM r2.1

Channel-mix decoder fine-tune per D-010 Track 1 (G4 zero-shot bake-off triggered the fine-tune-first pivot). NeurIPS 2024 release. Sub-1M parameters. Aggregate raw 50 Hz telemetry to 1 Hz mini-sector tensors before forecast.

LAYER 2 · PROJECTOR

Differentiable Cvxpylayers physics

Constant-mu friction-ellipse projection at HEAD. 8-tier Pacejka linearization + 3-iteration SCP outer-loop + jerk-bound enforcement are named swap-points per D-031 staged ladder. V1 NumPy validator reads coa_overlap_flag + routes the constraint per-driver.

LAYER 3 · AUDITOR

Granite Guardian 4.1 8B

BYOC custom-rule registry audits every projection log + every coaching narration. Server-side regex scrubber pre-pass + bounded retry-loop on the narrator path emitting retry directives on detected violations.

Two jobs split between two models. TTM forecasts the next-session pace envelope. Granite 4.1 8B Instruct turns the envelope plus the COA plus the debrief into a precision tuning recommendation. Aggregation done honestly.

COA-parameterized *simultaneity gate*.

MOAT 01

First on adaptive telemetry

First application of a pretrained time-series foundation model to adaptive motorsport telemetry.

MOAT 02

First COA as tensor input

First public AI race-engineer workflow we found that reads the FIA Certificate of Adaptations as a binding regulatory input.

MOAT 03

First simultaneity gate

First COA-parameterized brake-throttle simultaneity gate. coa_overlap_flag = 1 returns feasible. coa_overlap_flag = 0 returns violation. Same physical event, different verdict.

Standard tools assume able-bodied physics. They encode throttle × brake = 0. Sarah's electronic hand-control system has approved hardware that permits simultaneous brake and throttle inputs mid-corner. Her COA permits it. APEX recognises the input. We coach disabled drivers with the model that matches their car.

Sixty seconds. *Provenance on every line.*

Coaching report

Corner-by-corner cause + consequences + recommendation + evidence. 4-block CornerCard per D-058 reasoning_chain.

Tuning delta

Reduce hand-lever brake travel by 4 mm citing the exact COA section. From 38 mm to 34 mm. Section 3(c) hardware spec.

Forecast envelope

Next-session pace envelope across 30 mini-sectors. Mean line + confidence band. Granite TimeSeries TTM r2.1 channel-mix decoder.

Guardian verdict

BYOC custom-rule registry verdict + reasoning trace + reproducibility footer with model versions + input file SHA-256 hashes + audit-id.

> Every claim cites a specific Certificate of Adaptations section and a specific FIA Appendix L provision. HARD-COMPLIANCE server-side scrubber strips invented regulatory anchors. Self-Correcting Retry Loop emits retry directives on detected violations.

Fourteen tools. *Per-tool honesty tier.*

2 WIRED at HEAD + 10 INTEGRATION with named swap-points + 2 ACCELERATOR build-time. Per `app/frontend/lib/ibm-stack.ts`.

Granite Instruct 4.1 8B	WIRED
Granite 4.0 Nano 350M	WIRED · WEBGPU EDGE
Granite-Docling 258M	INTEGRATION
Granite Vision 4.1 4B	INTEGRATION
Granite TimeSeries TTM r2.1	INTEGRATION
Granite FlowState r1.1	INTEGRATION
IBM TSPulse 1M	INTEGRATION
Granite Embedding R2	INTEGRATION
Granite Guardian 4.1 8B	INTEGRATION · BYOC
Granite Instruct 4.1 3B	INTEGRATION · CHAT-ROUTING
Granite Speech 4.1 2B-Plus	INTEGRATION · STT
LangGraph + MCP Gateway + ContextForge	INTEGRATION · ORCHESTRATION
Docling library	ACCELERATOR · BUILD-TIME
Mellea v0.5.0	ACCELERATOR · IVR-LOOP SLOT

Each Granite tool does one thing in one place. Inspired by IBM's publicly documented Ferrari watsonx case study, redeployed on a safety-critical sensor-data domain.

Seven moves. *Three shipped, four scaffolded.*

01 · Constant-mu friction-ellipse projection

V2 Cvxpylayers ceiling at HEAD. Differentiable. Single-iterate.

02 · COA-parameterized gate

V1 NumPy validator reads coa_overlap_flag. Routes constraint per-driver.

03 · Byte-equality serializer

D-050 contract. V1 NumPy + V2 Cvxpylayers emit byte-identical violation strings modulo engine header.

04 · Pacejka linearization · Scaffold

8-tier projector swap-point V12 per D-031.

05 · SCP outer-loop · Scaffold

3-iteration unrolled SCP wrapper V13. ||grad L|| = 24.12 + FCVR = 0 on D-027 Stage C PASS.

06 · Mellea tri-agent critic · Scaffold

Physics + Pedagogy + Guardian-Safety critics. ACCELERATOR tier; flips when M3-V5.2 lands.

Plus IBM TSPulse polyphase anomaly endpoint + Granite TimeSeries TTM in-browser scaffold. All scaffolded with named swap-points. Three-layer pipeline above stands on the three shipped.

Composite *A*.

AXIS 01

Technical Execution

A

14 IBM Granite tools · 196 backend tests · 65+ frontend vitest · Playwright fidelity walks · Vercel production deploy · Apache 2.0 public from inception.

AXIS 02

Innovation

A

COA-parameterized simultaneity gate killshot · frozen-TSFM + differentiable-physics-projection composition · byte-equality serializer regression contract · APEX-Bench v0.1.0 public LIPS leaderboard · NeurIPS Workshop paper draft in repo.

AXIS 03

Challenge Fit

A+

Adaptive-racer + veteran-team-driver + grassroots-competitor tri-persona ladder · MME Motorsport per-surface consent · Mission 44 outreach · Team BRIT engineering correspondence · ISO 26262 vocabulary alignment per paper section 3.8.

AXIS 04

Implementation & Feasibility

A

19/19 production routes respond 200 · APEX-Bench public leaderboard reproducible from clone · named swap-points for every INTEGRATION tier tool · Vinh-side Phase 6 backend gates G6 G7 G8 G9 G10 all PASS.

CLOSE

The race engineer for the drivers who *do not have one*.

APEX is a reference architecture for governed foundation-model deployment on safety-critical sensor data. Adaptive racers, veteran-team drivers, grassroots competitors. Public from Day 1.

DEMO
apex-one-black.vercel.app

SOURCE
github.com/StephenSook/apex

LICENSE
Apache 2.0

Stephen Sookra & Vinh Le · Kennesaw State University · 12 days